P6: Probability & Bayes' Rule

FIT1061 Pass Task — Week 6 — The Confidence Machine

Probability is a core tool in AI because it gives us a language for uncertainty. In this task, the emphasis is on understanding how evidence changes belief and how a simple probabilistic model can support decision making under uncertainty.

Show us that you can

- You can reason about probability clearly.
- You can explain how Bayes' rule updates belief.
- You can interpret the result, not just calculate it.

Give us

- A photographed hand-computation of Bayes' rule on YOUR personalised scenario (P6.1) — your single appended file.
- A completed probability notebook with Bayes' rule calculations on the spam dataset and COVID example, with your reflection written inside it (Part D).

Resources

- Week 6 lecture slides and recording *(Moodle)*
- P6 starter notebook *(OnTrack)*
- Week 6 lecture notes — Probability & worked examples *(Moodle)*

Key sections in the task notebook

- Part A: Building a spam detector from raw data
- Part 0: Hand-compute Bayes on YOUR scenario (P6.1)
- Part B: The base-rate trap (COVID testing)
- Part C: Bridge to Naive Bayes (preview of Week 7)
- Part D: Reflection — "What goes wrong when we deploy probabilistic AI in high-stakes domains?" (P6.2)

Do the following

1. Read the lecture material and make sure you understand the basic probability setup.
2. Complete the notebook or worksheet in order.
3. Explain each step in your own words, especially the Bayesian update.
4. Upload your completed notebook plus a single photo/PDF of your hand computation; your reflection stays inside the notebook.

! Acceptable use of Artificial Intelligence

AI tools are **discouraged** for implementation tasks at the Pass level. You may use AI for *explanation* (e.g., "explain what a queue is"), but not to generate code or write your reflection. Cite all AI use. The milestone discussion will probe for genuine understanding.
All submitted work must be your own.